that the other ports are, so that when the piston is at the upper end of the compression stroke the valve is opened and the first gas enters the crank case in the same manner as in position A, Figure 2. Aside from this the operations are exactly the same as in the two-port cycle.

These engines may be reversed by changing the time of ignition. Other advantages are simplicity, light weight, and cheapness. Some of its disadvantages are that it never has a clean mixture: always some of the burned gases are mixed in with the fresh charge. But in well designed engines the amount of this mixture is not serious. The explosions being more frequent, the cooling and lubricating problems are necessarily harder.

For general farm use the four-cycle is to be recommended as being more reliable and more economical in the long run.

The Details of a Gas Engine.

The elements and principal events of gas engine operation have already been cited, and it will be unnecessary to repeat them here. A study in detail of the parts and process that make the gas engine a success will be taken up.

Carburetion.

As previously mentioned, whether gas or liquids are burned in the cylinder, it must be properly prepared first. The liquid must be vaporized and the proper amount of air added to support the combustion. In the gas fuel only the mixture of air need be added. There are numerous devices for receiving the gas, or gasoline or other liquid fuel vaporizing and mixing the vapor with air. A carburetor is a mixer of fuel with air. On many engines the carburetor is called a mixer. See Figure 3.

When using a liquid fuel it must be broken up into fine particles or vaporized. Secondly, it must be thoroughly mixed. Thirdly, the proper amount of air must be added.

The operation. mixing is a simple process. The accompanying illustration will show one way in which it is done. The large pipe is the air inlet, and the small one meeting it at right angles is the fuel supply. When the air is drawn into the cylinder the fuel is drawn into the passageway by the same suction, and as it passes the needle valve it is divided into a thin spray. The air picks it up and carries it into the cylinder. The needle valve not only breaks up the fuel into a spray, but by its adjustment regulates the amount of fuel mixed with the air to make a perfect combustion.

Most of the air passageway is obstructed at the part where the fuel enters, in order to increase the velocity of the air as it passes the fuel inlet, which helps to break it up into a finer spray or vapor.

The liquid fuels which are on the market today do not completely vaporize by the above proces Many methods to accomplish this have been invented and put on the market. Since gasoline has become so high priced, because of its scarcity, the manufacturers, in order to get more of the product have

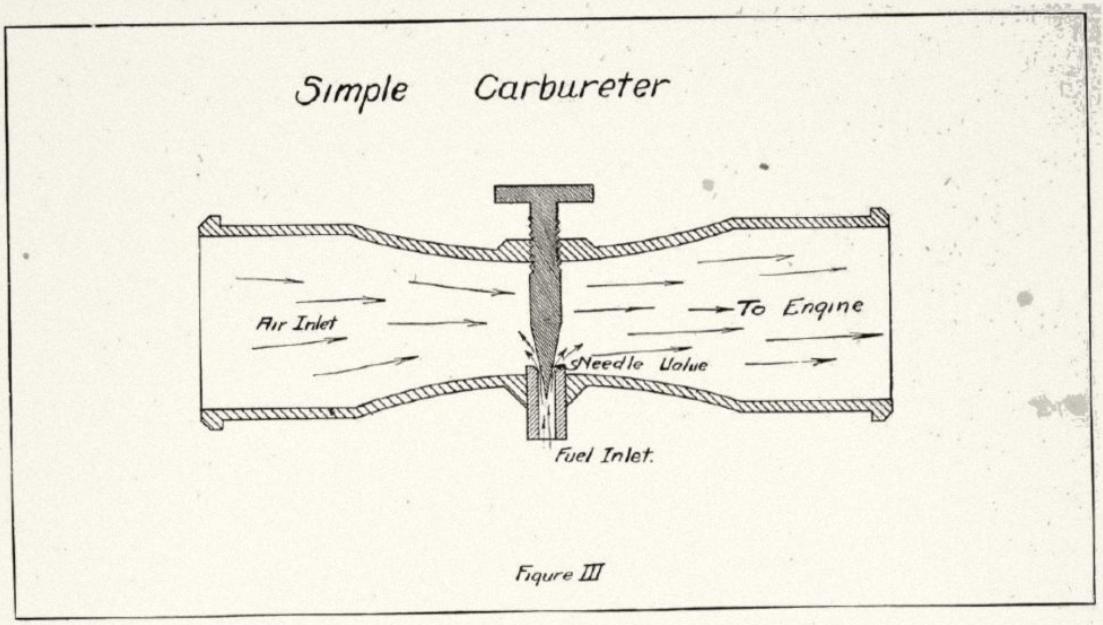

Simple Carbureter
Air Inlet
To Engine
Needle Valve
Fuel Inlet.

Figure III

decreased its quality. This lower grade of gasoline is harder to vaporize than the original gasoline used when automobiles and gas engines first came into use.

Various methods of applying heat to the gas are now on the market and one of the simplest schemes is to take the air which is used in the mixture from off the hot pipes that leave the exhaust chamber. After the engine is running and the exhaust has heated the pipes, this method works all right, but in cold weather it is hard to start such a device until the carburetor and adjacent parts are warm. This can be done with hot cloths or hot water.

Adjusting the Mixture of Fuel and Air.

As the supply of air for the mixture is drawn from the atmosphere, it will be subject to the change of atmospheric conditions.,A change in temperature will change the volume admitted and also the amount of moisture that may be carried by the air. It will also change the vaporizing power as noted above. Therefore carburetors are fitted with certain adjustments to change the proportion of fuel and air to meet atmospheric conditions. A simple carburetor is shown in Figure III.

Also the speed of the engine will vary the suctions so that adjustments must be made automatic to give the proper mixture of fuel during high speed as well as low. The best carburetors on the market will do this.

It is the function of the carburetor or mixer to properly proportion the gases before they are turned into the engine. To regulate the speed of the engine, the governor will open and close the throttle as the speed is being increased or decreased respectively. The throttle is nothing more than a damper valve placed in the passageway between the carburetor and the cylinders. In the hit and miss type of governor, the number of charges taken into the cylinder, is controlled to meet variations in load conditions.

By experiment, it has been found that there is one mixture of air and fuel which is the most economical, i. e., one which will produce perfect combustion and give out the most power. If more fuel is used, the mixture is too rich and will not burn properly, if less fuel is used, the mixture is too lean with practically the same result. Either too much or too little is a waste.

Lean mixtures will cause flames to pass through the air openings, which is known as backfiring. Neither will the engine pick up speed when the mixture is too lean, for it lacks power.

Back-Firing.

Back-firing is caused by a lean mixture as explained above. The fuel charge burns so slowly in the cylinder that when the intake valve opens on the following suction stroke, the gas which is still flaming ignites the new charge, which explodes back through the carburetor; through the air passage.

To remedy this trouble, open the needle valve to give more fuel or cut down the amount of air, which is the same thing. The manufacturer of

every gas engine will give proper instructions for the regulating or adjusting of the carburetor for the conditions which are liable to happen during its life.

Rich Mixture and Its Effect.

When black smoke issues from the exhaust pipe a rich mixture is indicated. An odor of fresh fuel will also be detected in the waste gases. This waste fuel ignites and explodes in the exhaust passages and sometimes blows off the muffler. A rich mixture will also heat up the cylinder quickly, and the ignition points or spark plugs will become covered with carbon deposits in a short time and will make the cylinders miss, a decrease in the amount of power resulting.

To correct this, the process is just opposite to a lean mixture—either close the fuel valve or add more air.

When the mixture is properly proportioned, the exhaust will be clear. Do not mistake the blue smoke coming from the exhaust for a rich mixture, because this is due to over-lubrication. It is a black smoke which indicates too rich a mixture. In kerosene engines a blue smoke may be seen coming from the exhaust at times. This will indicate incomplete combustion or excess lubrication.

Conclusion—The method of the modern manufacturer is to test out the mixing values and carburetors of his special engine and if the fuels which he recommends are used, there should be no trouble in operating the engine. Study carefully the discussion given here, and most cases of trouble will be taken care of with the least expense.

Gas Engine Combustion.

"The ordinary definition of "combustion" means simply burning. The burning of fuel in the gas engine cylinder is the same simple chemical process. It takes place under peculiar conditions. Ordinarily one sees these same fuels burning in the open with excess of air. But actually the fuel is in excess and the burning smokes considerably. In a gas engine cylinder, the mixture of fuel and air is so perfectly made that the result of the combustion is a clear gas and the time of burning is so sudden that it is called an explosion.

Fuels.

All products of petroleum, such as gasoline, kerosene, etc., are composed of the compounds of hydrocarbons which are combinations of carbon and hydrogen., These two elements occur in various proportions depending upon. the specific gravity and the addition of other elements. Atmospheric air is a mixture of, approximately, 23 parts by weight of oxygen and 77 parts by weight of nitrogen. This is a physical mixture.

Carbon and hydrogen as a fuel are in chemical combination and in order for combustion to take place the combination must be broken up. It is partly accomplished by the spray, and partly by intense heat.

THE COMBUSTION.

The air is our chief supply of oxygen. Oxygen is necessary to support combustion. This is why air and the fuel are drawn into the cylinder together. As soon as they are brought into the cylinder, they are compressed, in order to bring the molecules of the oxygen and of the hydrocarbons close together. Then by electric spark or some flame the mixture is ignited and an intense heat occurs as a result of the combustion of all these elements. The carbon and oxygen unite to make carbon dioxide, and the hydrogen unites with the oxygen to make water.

It would be ideal if only oxygen could be added to the fuel mixture, but it is cheaper to get the oxygen from the air, and since the air contains only 23 percent oxygen, a large quantity of it must be used. It has been theoretically determined that 203 cu. ft. of air are required to burn one pound of gasoline. In practice, from 240 to 260 cu. ft. are drawn into the cylinder for the proper combustion of the pound of fuel. It has been said before that each engine has its own peculiarities. The amount of air and gasoline should be mixed according to its demand.

Combustion and Explosion.

The process of combustion in the engine is oftentimes called an explosion. This needs an explanation. It is very near an explosion, and in one sense it can be said that itis. An explosive compound is a mixture of carbon, hydrogen and oxygen chemically, so that there is no necessity for an outside supply of oxygen, that is, all of the combustible parts are contained within the chemical mixture or compound. When set on fire the chemical change takes place without the assistance of any outside influence excepting heat. Now the difference is only this: hydrocarbons of the fuel in the gas engine are mixed physically with the oxygen of the air, while the hydrocarbons in an explosive compound are combined chemically with the oxygen. The explosion is quicker, more certain and reliable. The combustion in the gas engine cylinder is slower!

Timing the Ignition.

The compression does not take place instantly. The ignition of a charge is usually a little bit previous to the time when it is actually needed in order to give the flame a chance to spread throughout the whole mixture. Therefore, the spark is applied to the mixture a little bit before the piston arrives at the compression end of the cylinder. As soon as it has completed its motion, combustion will have been completed, the temperature of the gases raised to the highest point, and the pressure developed so that the piston will be pushed back with maximum speed.

To the human senses it appears that combustion takes place instantaneously. However, with accurate instruments it will be found that this is not true. Some fuels will burn faster than others, and the speed is also affected by the volume of the charge, the temperature, and the perfection of the vaporization and proportioning of the air. Atmospheric changes during the year